LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424055
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Day	Tuesday

SET 20: Stock Analysis

Aim

To analyze stock prices using a line chart, daily percentage-change bar chart, data table, and an interactive Tableau dashboard.

Dataset: Stock Prices

The provided data is slightly misformatted. Interpreting it as three dates with prices for Stocks A, B, and C gives:

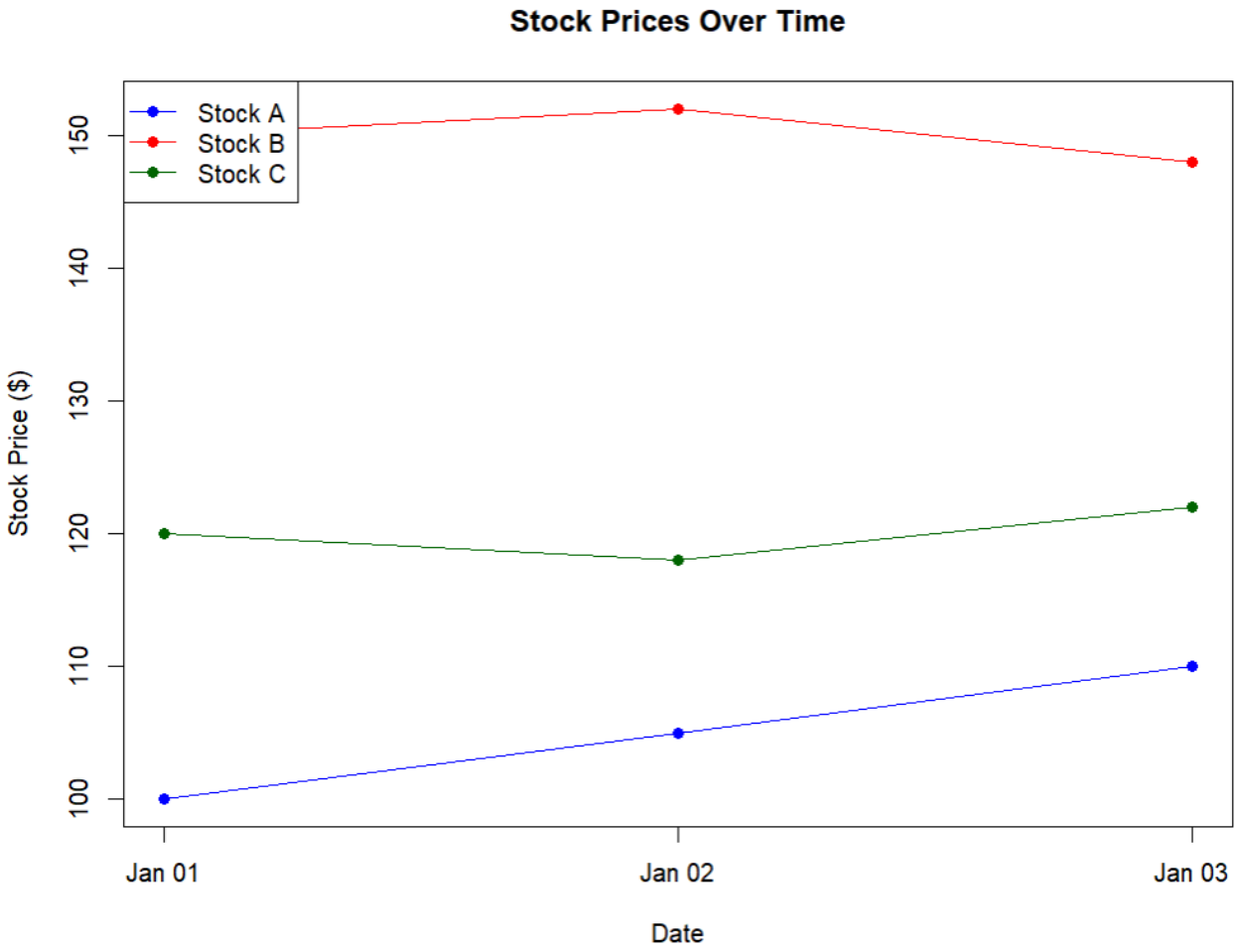
Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Q1. Create a line chart to visualize the stock prices of three companies over time.

R Program

```
date <- as.Date(c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03"
))
stock_A <- c(100, 105, 110)
stock_B <- c(150, 152, 148)
stock_C <- c(120, 118, 122)
plot(
  date, stock_A,
  type = "o",
  pch = 19,
  col = "blue",
  ylim = range(c(stock_A, stock_B, stock_C)),
  xlab = "Date",
  ylab = "Stock Price ($)",
  main = "Stock Prices Over Time"
)
lines(date, stock_B, type = "o", pch = 19, col = "red")
lines(date, stock_C, type = "o", pch = 19, col = "darkgreen")
legend(
  "topleft",
  legend = c("Stock A", "Stock B", "Stock C"),
  col = c("blue", "red", "darkgreen"),
  lty = 1,
  pch = 19
)
```

Output:



Observation: Stock A increases from 100 to 110, Stock B rises slightly and then falls to 148, and Stock C falls before rising to 122.

Q2. Generate a bar chart showing the daily percentage change in Stock A.

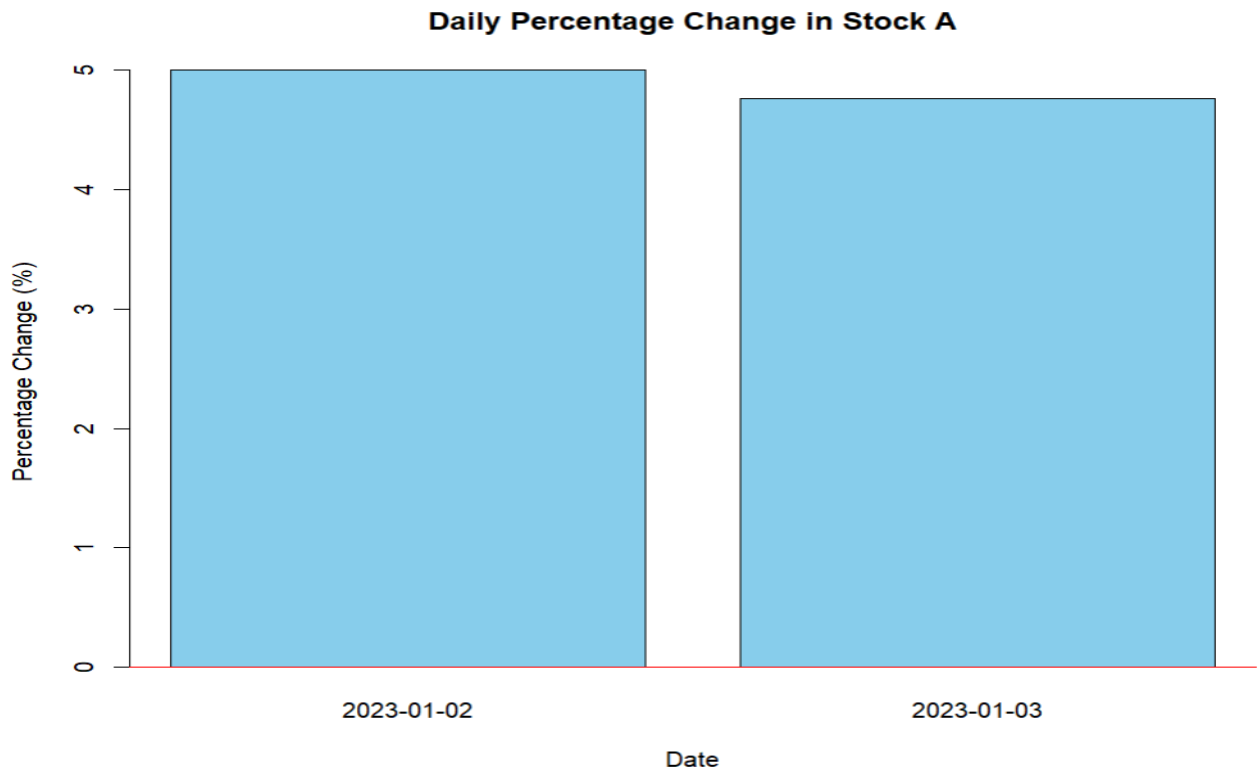
Formula:

$$PercentageChange = \frac{CurrentPrice - PreviousPrice}{PreviousPrice} \times 100$$

R Program

```
stock_A <- c(100, 105, 110)
percentage_change <- diff(stock_A) / head(stock_A, -1) * 100
change_dates <- c("2023-01-02", "2023-01-03")
barplot(
  percentage_change,
  names.arg = change_dates,
  xlab = "Date",
  ylab = "Percentage Change (%)",
  main = "Daily Percentage Change in Stock A",
  col = "skyblue"
)
abline(h = 0, col = "red")
```

Output



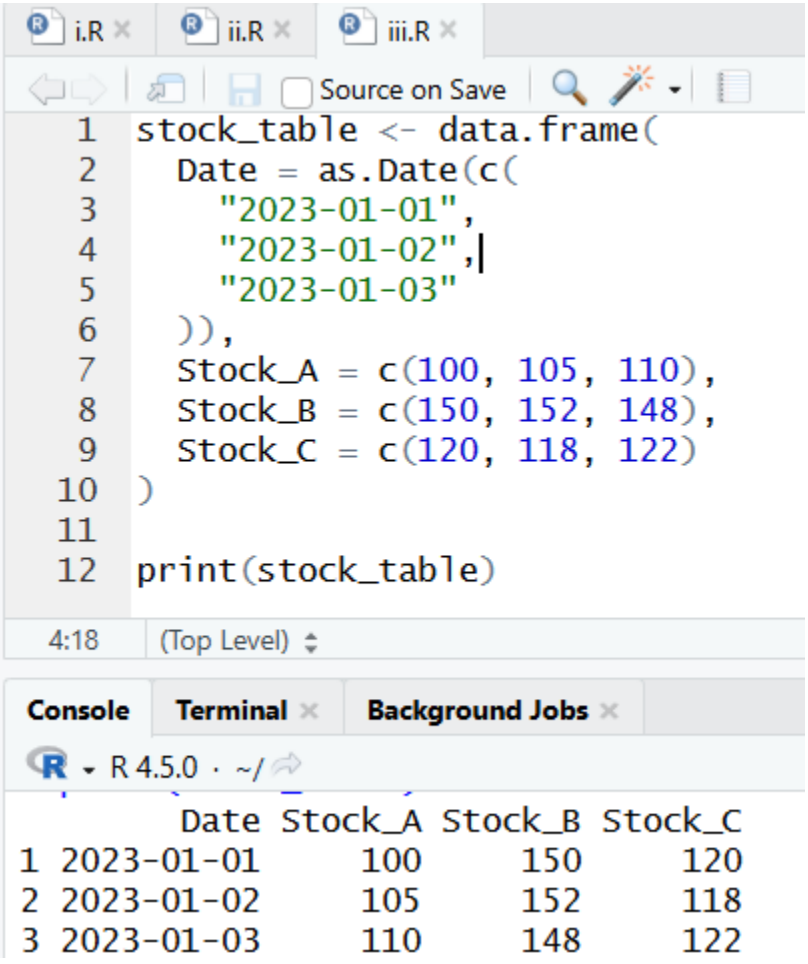
Observation: Stock A increased on both days, with a slightly smaller percentage gain on January 3.

Q3. Build a table to display the stock price data for each company.

R Program

```
stock_table <- data.frame(
  Date = as.Date(c(
    "2023-01-01",
    "2023-01-02",
    "2023-01-03"
  )),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)
print(stock_table)
```

Output



Observation: The table displays the daily prices of all three stocks for the given period.

Q4. Develop a Tableau dashboard combining the line chart, bar chart, and table.

Step 1 — Prepare the CSV dataset

Save this content as Stock_Prices.csv.

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Import Stock_Prices.csv.
4. Ensure Date is recognized as a Date field.

Step 3 — Create the line chart

1. Open a new worksheet named Stock Price Trend.
2. Drag Date → Columns.
3. Select the date level Day if needed.
4. Drag Stock A, Stock B, and Stock C to Rows.
5. Alternatively, pivot the three stock columns into Stock and Price, then drag Date → Columns, Price → Rows, and Stock → Color.
6. Choose Line from the Marks dropdown.
7. Set the title to Stock Prices Over Time.

Step 4 — Create the daily percentage-change bar chart

1. Create a new worksheet named Stock A Daily Change.
2. In the Data pane, select Create Calculated Field.
3. Name the field Daily Percentage Change.
4. Use the formula below, assuming the stock data has been pivoted into Stock and Price fields

LOOKUP(SUM([Price]), -1) * 100

5. Drag Date → Columns and select Day.
6. Drag Daily Percentage Change → Rows.
7. Drag Stock → Filters and select Stock A.
8. Select Bar from the Marks dropdown.
9. Right-click the calculated field and select Edit Table Calculation. Set Compute Using → Date.
10. Set the title to Daily Percentage Change in Stock A.

The first date has no previous day's price, so its percentage change will be undefined. The other two dates should show approximately 5.00% and 4.76%.

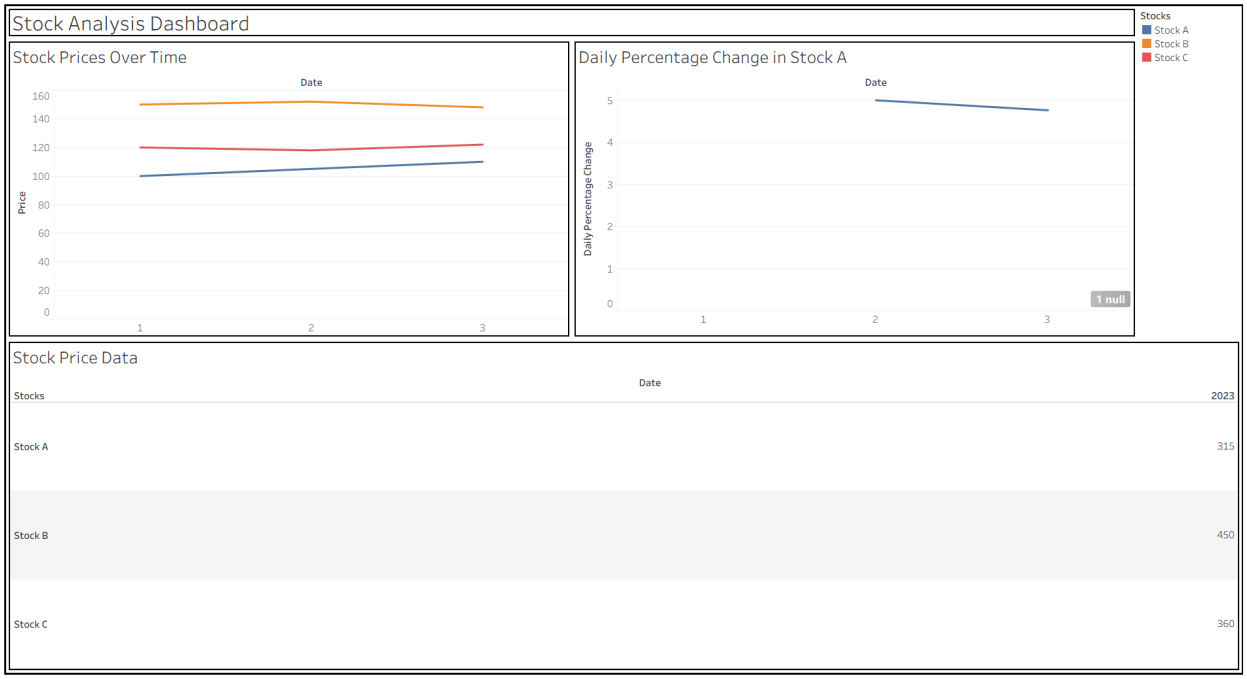
Step 5 — Create the stock price table

1. Create a new worksheet named Stock Price Table.
2. Drag Date → Rows.
3. Add Stock A, Stock B, and Stock C as displayed measures.
4. Choose Show Me → Text Table, if required.
5. Set the title to Stock Price Data.

Step 6 — Create the dashboard

1. Click New Dashboard.
2. Add the Stock Price Trend line chart.
3. Add the Stock A Daily Change bar chart.
4. Add the Stock Price Table.
5. Arrange the two charts at the top and the table below.
6. Add date filters if required.
7. Set the dashboard title to Stock Analysis Dashboard.

Expected Dashboard



Procedure

1. Prepared the stock price dataset for three companies.
2. Created a line chart to compare stock prices.
3. Calculated Stock A's daily percentage changes and plotted a bar chart.
4. Built a table containing all stock prices.
5. Combined the visualizations in a Tableau dashboard.

Result

The stock price data was analyzed using a line chart, daily percentage-change bar chart, and table. A Tableau dashboard was designed to support interactive exploration of stock prices.